

<!-- multilingual.md – exercises three orthogonal axes of non-English text rendering in mdlout/Lout:

- Latin / Western-European prose with accented characters. mldout's markdown source is UTF-8, but the Lout binary reads ISO-Latin-1, so UTF-8 multi-byte sequences round-trip incorrectly. The robust path is to drop into a raw-Lout fence and use @Char "eacute"/ "agrave" etc., or the bsf shorthands "" "" "-" "_" "...". Both back-ends honour this route.
- Greek letters via the Adobe Symbol font (@Sym alpha, @Sym Pi, ...). Recent z53.c work (commit ec987be) wired the Symbol glyph table into the SVG back-end, so HTML output now matches the PostScript reference glyph-for-glyph.
- A KaTeX math equation that itself uses Greek + the Symbol math operators. Renders fully in HTML; PDF mode falls back to the @Math placeholder.

A small Cyrillic specimen at the bottom shows how to invoke Lout's built-in Russian language pack via raw-Lout passthrough; see `cookbook.md` recipe 18 for the gotcha discussion. →

A Multilingual Sampler

This page exercises m^dlout's handling of three character regimes in one document: accented Latin glyphs that ride on the document's base font, Greek letters drawn from the Adobe Symbol font via @Sym, and a $\$...$ / \$\$...\$$ math block whose LaTeX commands include further Greek and math operators.

Latin: accented Western-European text

Lout's standard ISO-Latin-1 character map covers the routine Western-European accent menagerie. Because mdlout reads UTF-8 markdown but Lout reads ISO-Latin-1, the safest path for accented letters is to drop into a raw-Lout fence and request each glyph by Adobe PostScript name with @Char:

Le caf é – la place É toile – na ĩ ve M ü nchner – cr ê pes Suzette – l' â ne – El Ni ñ o – S ã o Paulo

Punctuation has shorthands in bsf: ‘and”render directional quotes,—is an en-dash,—an em-dash,...ellipsis. Those work inline without leaving markdown. A short worked sentence: this manual uses‘directional quotes” and the occasional em-dash—like so—in ordinary prose.

Greek: the Symbol font via @Sym

The Adobe Symbol font supplies a full lowercase + uppercase Greek alphabet plus the standard math operators. Lout exposes it through the `@Sym` macro in the `bsf` library. Recent `z53.c` work wired the Symbol glyph table through the SVG back-end, so the HTML output below is glyph-identical to the PostScript reference.

$\alpha \quad \beta \quad \gamma \quad \delta \quad \varepsilon \quad \zeta$
 $\eta \quad \theta \quad \iota \quad \kappa \quad \lambda \quad \mu$
 $\nu \quad \xi \quad \pi \quad \rho \quad \sigma \quad \tau$
 $\phi \quad \chi \quad \psi \quad \omega$

$\text{\AA} \quad \text{\AA} \quad \Gamma \quad \Delta \quad \text{\AA} \quad \text{\AA} \quad \text{\AA} \quad \text{\AA} \quad \text{\AA} \quad \text{\AA}$
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A handful of Symbol operators used in scientific text: integration, summation, square root, infinity, partial derivative, gradient, and the usual plus-or-minus / multiply / less-or-equal / greater-or-equal / set-membership glyphs.

$\int \quad \sum \quad \sqrt{\quad} \quad \infty \quad \quad \partial \quad \nabla \quad \pm \quad \times \quad \quad \leq \quad \geq \quad \in \quad \rightarrow$

Math: a Greek-laden equation block

KaTeX handles the LaTeX command set in HTML mode; in PDF mode the same block falls back to the `@Math` placeholder. The Euler identity, the golden ratio, and an integral over the real line:

$e^{i\pi} + 1 = 0, \quad \varphi = \frac{1 + \sqrt{5}}{2}, \quad \int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi}.$

A second display, showing the Riemann zeta function at $s = 2$ and the Gauss-Bonnet formula:

$\zeta(2) = \sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}, \quad \int_M K, dA + \int_{\partial M} k_g, ds = 2\pi \chi(M).$

Inline math also flows through: the fine-structure constant $\alpha \approx 1/137$, the Planck relation $E = h\nu$, and the spin-half operator commutator $[\sigma_x, \sigma_y] = 2i\sigma_z$.

Cyrillic: a Russian-language paragraph

Lout ships a Russian language pack (`@SysInclude { russian }`) that wires KOI8-R input through the Russian PostScript fonts. Driving it from the markdown layer is a two-step dance: switch the document language inside a raw-Lout fence with `{ Russian } @Language`, then type the Cyrillic text inside that fence. A Roman-letter transliteration is shown here because the SVG back-end does not yet ship the Cyrillic glyph table.

Privet, mir! – a Roman-letter transliteration.

The `Russian` package itself is the reference for what genuine Cyrillic input looks like; see `lout/include/russian` and the `KOI8-R.LCM` character map in `lout/maps/`. Wiring full UTF-8 Cyrillic through `z53.c` is tracked in `TODO.md`.

Closing observation

Three writing systems, one Lout pass. The PDF pipeline (`mdlout -format=pdf`) is the canonical archival format for the Latin and Greek sections; HTML is the canonical format for the math (KaTeX) and for copy-and-paste fidelity of accented characters.